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Takada et al.

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(54) **STORAGE DEVICE AND A METHOD OF ERROR PROCESSING THEREOF**

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CPC **G06F 11/1064** (2013.01); **G06F 12/0871**
(2013.01); **G06F 12/126** (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

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(57)

ABSTRACT

In a storage device including a plurality of controllers for controlling data reading and writing operations with respect to at least one host computer, each of the controllers includes: a cache region to which a plurality of management units capable of temporarily storing the data are allocated in accordance with the data reading and writing operations; and a control unit for controlling the data reading and writing operations, and when an error occurs, the control unit determines whether or not an occurrence position of the error is in the cache region, and when it is determined that the occurrence position is in the cache region, the control unit excludes, among the plural management units, a specific management unit that contains the occurrence position from an allocation target in the cache region, to control the data reading and writing operations using remaining management units in the plural management units.

10 Claims, 8 Drawing Sheets

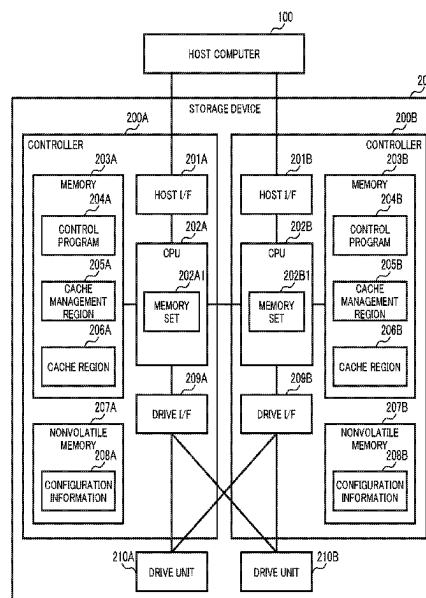


FIG. 1

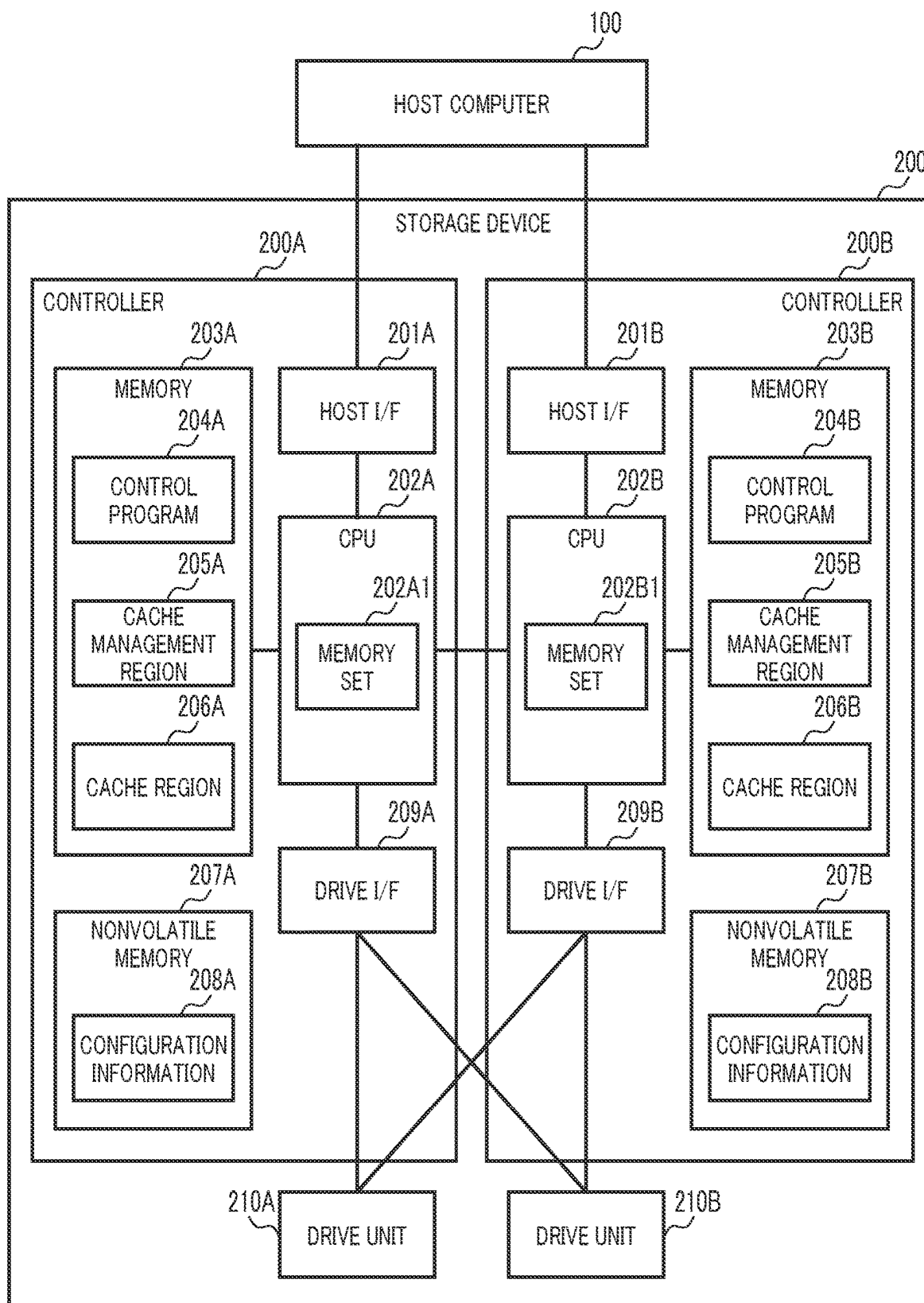


FIG. 2

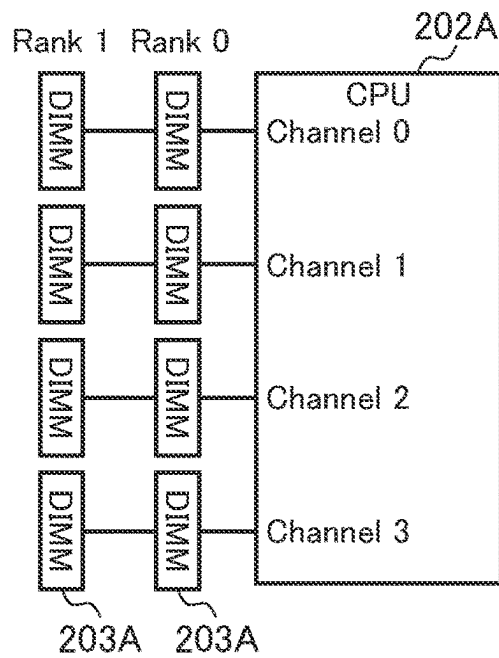


FIG. 3

ITEM	SET VALUE
Channel SELECTION BIT	6 – 7
Rank SELECTION BIT	8

FIG. 4

PHYSICAL ADDRESS

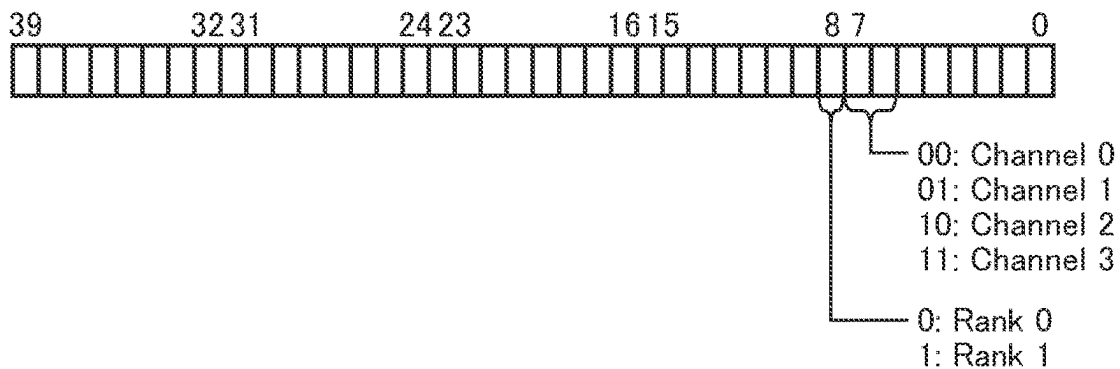


FIG. 5

211

HARDWARE FAILURE MANAGEMENT TABLE

ERROR NO.	ERROR OCCURENCE ADDRESS	OCCURRENCE OF UNCORRECTABLE ERROR	NUMBER OF CORRECTABLE ERROR OCCURRENCES
#0	0x012345640	Y	0
#1	0x2c003e4c0	N	10
#2	0c3f20e600	N	50

FIG. 6

212

REPLACEMENT TARGET MANAGEMENT TABLE

REPLACEMENT TARGET LIST	SERIAL NO.
DIMM (Ch 0, Rank1)	0x12345678
...	...

FIG. 7

CACHE DIRECTORY

213

VOLUME NO.	SEGMENT NO. IN VOLUME	SEGMENT NO. IN CACHE REGION	ATTRIBUTE
0	0	3	Clean
0	1	1	Dirty
0	...	...	...
1	0	4	Clean
1	1	—	—
1	...	...	...
...	...	...	...

FIG. 8

REVERSE LOOKUP TABLE

205C

SEGMENT NO. IN CACHE REGION	VOLUME NO.	SEGMENT NO. IN VOLUME
0	—	—
1	0	1
2	—	—
3	0	0
4	1	0
5	—	—
...	...	...

FIG. 9

FREE LIST

205D

NO. OF UNUSED SEGMENT IN CACHE REGION
0
5
...

FIG. 10

USAGE EXCLUSION LIST

205E

NO. OF UNUSABLE SEGMENT IN CACHE REGION
2
...

FIG. 11

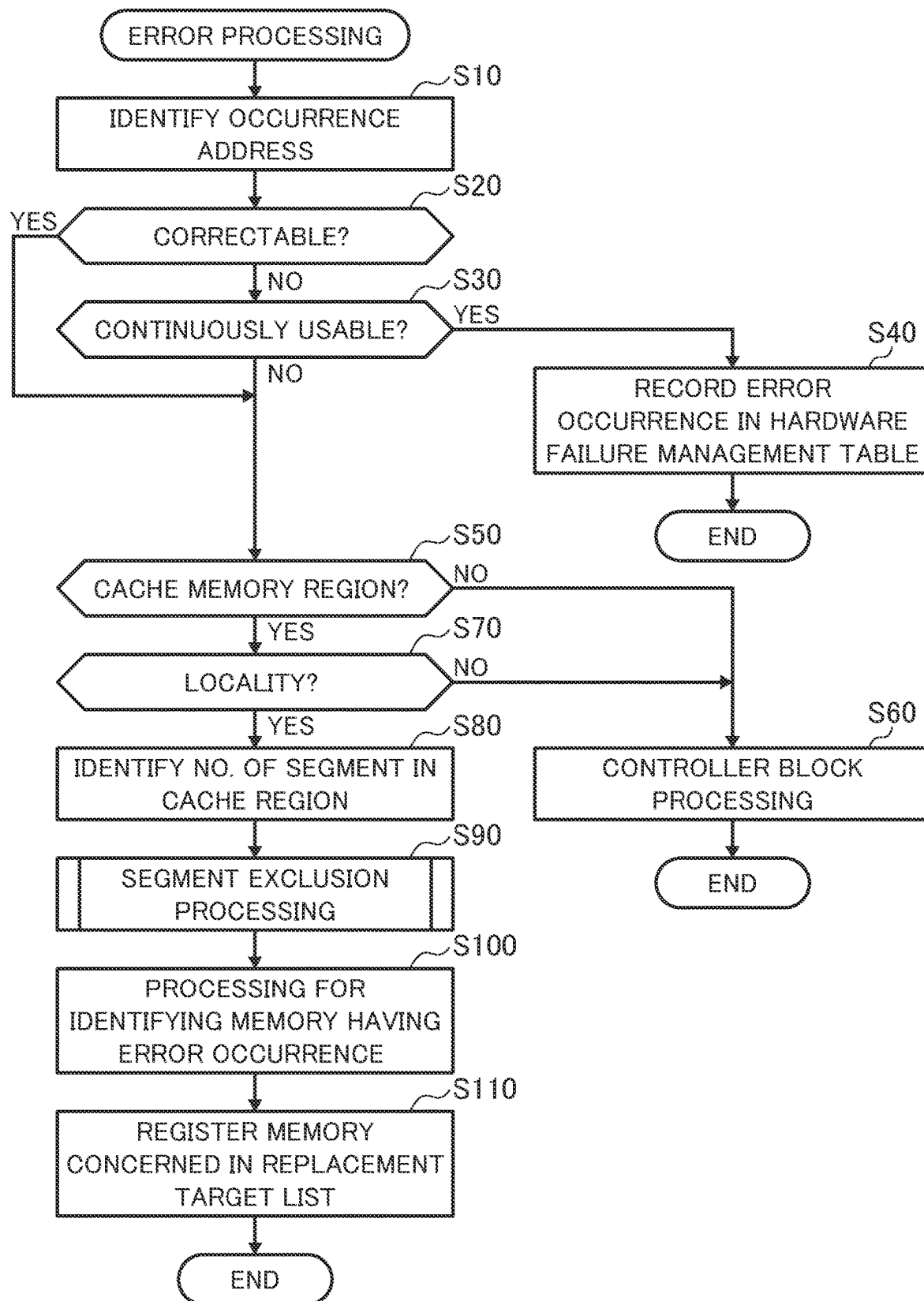


FIG. 12

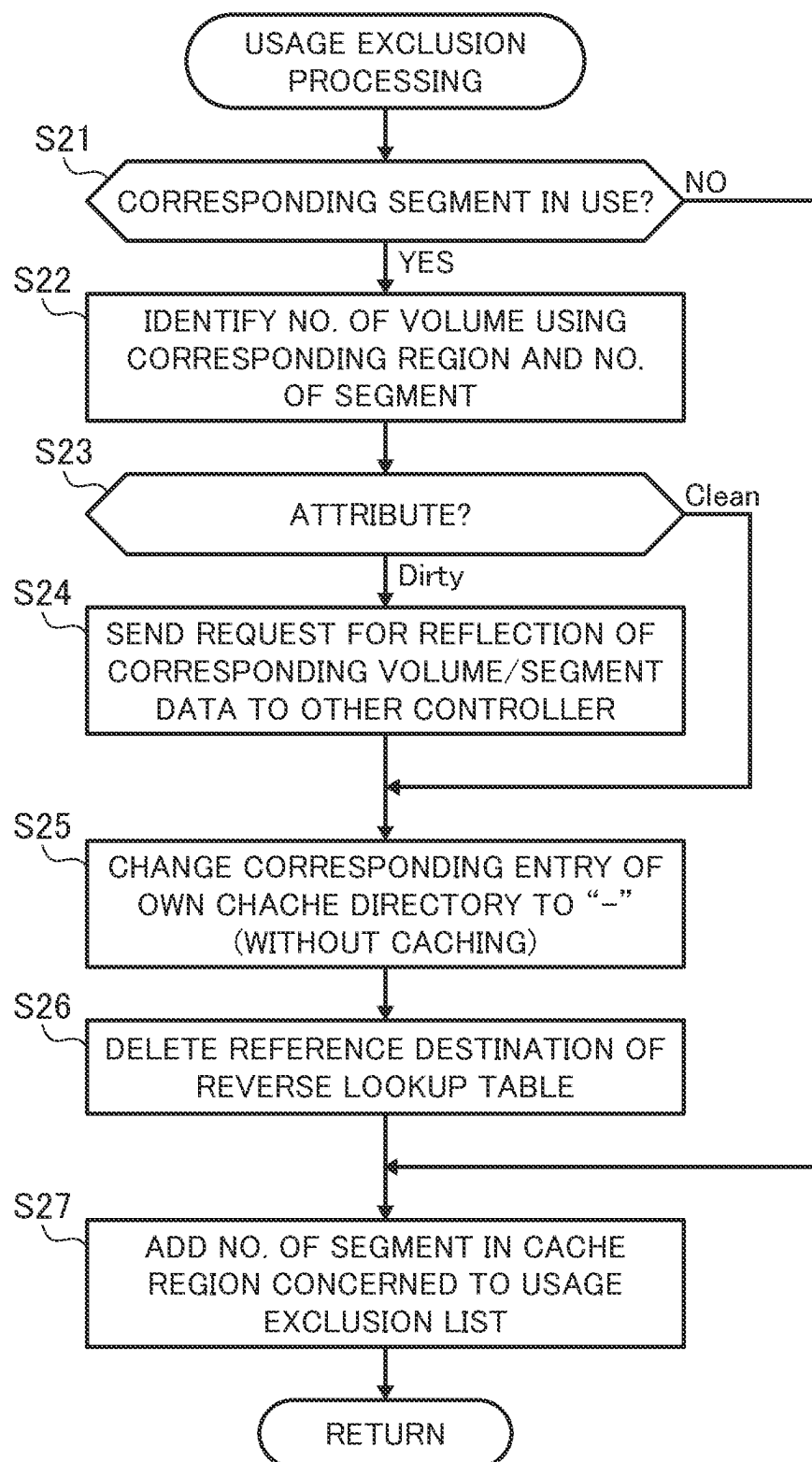
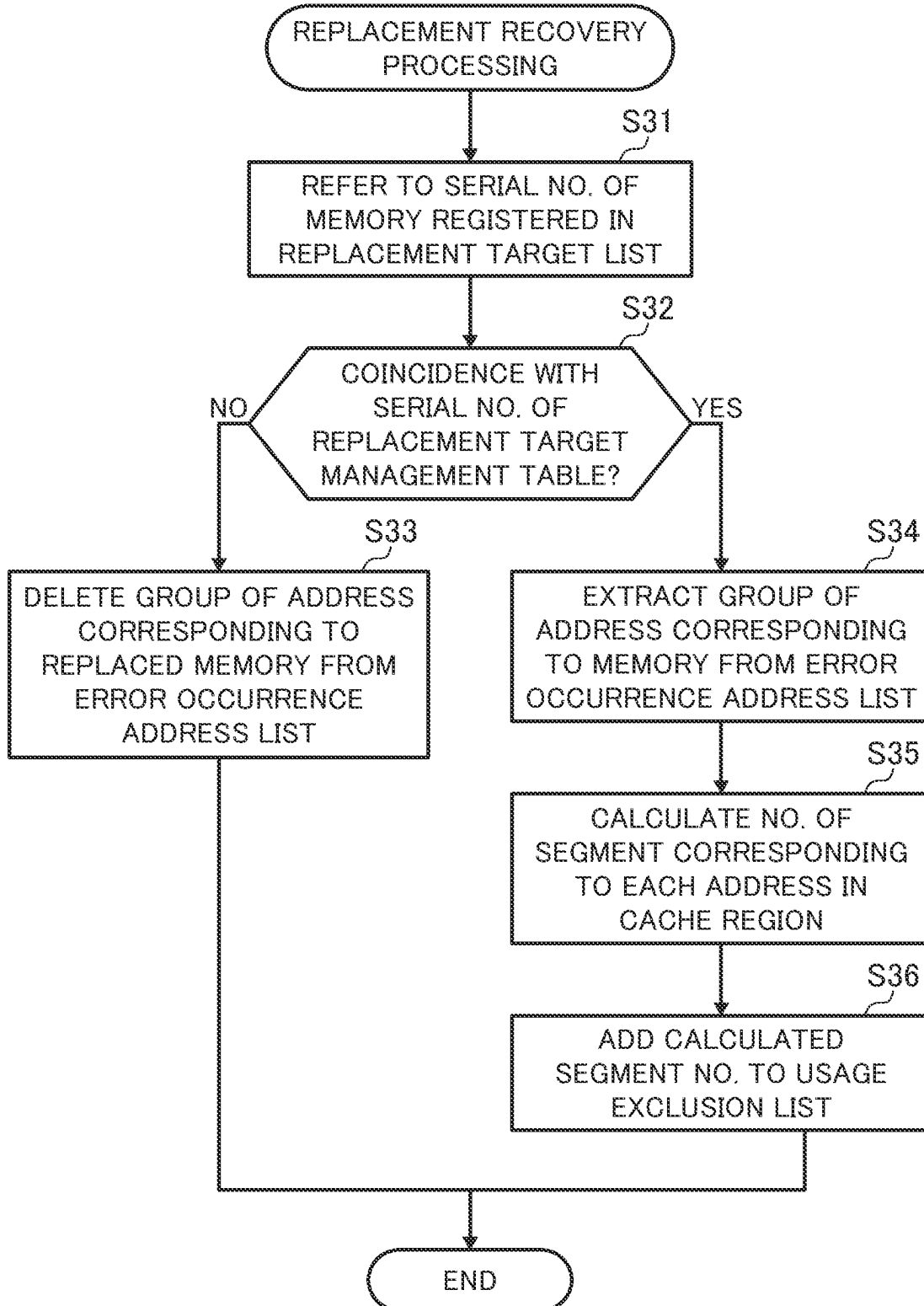


FIG. 13



STORAGE DEVICE AND A METHOD OF ERROR PROCESSING THEREOF

BACKGROUND

The present invention relates to a storage device and an error processing method. Preferably, the present invention is applicable to a storage device configured to prevent blocking of an entire controller despite an error that occurs in the cache region of the storage device.

A generally employed storage device has multiple controllers each provided with a memory and a cache region, and executes data reading/writing operations with respect to a host via a host I/F (Interface). The storage device is configured to duplex data in the memories and cache regions of the respective controllers. When an error occurs in the memory or the cache region of any one of the controllers, the controller having the error is blocked to allow the other controller to continue data reading/writing operations (refer to Japanese Unexamined Patent Application Publication No. 2004-199420).

SUMMARY

Upon occurrence of an error in the memory or the cache region of the controller, the storage device as disclosed in Japanese Unexamined Patent Application Publication No. 2004-199420 cannot continue the data reading/writing operations via the host I/F using the controller having the error. Accordingly, the storage device as a whole has to be operated in the state where the data duplexing is deteriorated.

The present invention has been made to propose a storage device and an error processing method to prevent blocking of the controller having an error which occurs in a cache region as much as possible.

Solution to Problem

In order to solve the above problems, the storage device comprises a plurality of controllers for controlling data reading and writing operations with respect to at least one host computer. In the storage device, each of the controllers includes: a cache region to which a plurality of management units capable of temporarily storing the data are allocated in accordance with the data reading and writing operations; and a control unit for controlling the data reading and writing operations, and when an error occurs, the control unit determines whether or not an occurrence position of the error is in the cache region, and when it is determined that the occurrence position is in the cache region, the control unit excludes, among the plural management units, a specific management unit that contains the occurrence position from an allocation target in the cache region, to control the data reading and writing operations using remaining management units in the plural management units.

Further, in the present invention, the error processing method for a storage device having a plurality of controllers each provided with a control unit for controlling data reading and writing operations with respect to at least one host computer, comprises: a management unit allocation step for causing the control unit to allocate a plurality of management units capable of temporarily storing the data to each cache region of the controllers in accordance with the data reading and writing operations; a determination step for causing the control unit to determine, when an error occurs, whether or not an occurrence position of the error is in the

cache region; and a control step for causing the control unit, when it is determined that the occurrence position is in the cache region, to exclude a specific management unit including the occurrence position among the management units from the allocation target in the cache region, and to control the data reading and writing operations using remaining management units in the plural management units.

According to the present invention, it is possible to prevent blocking of the controller having an error which occurs in a cache region as much as possible.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 mainly shows a system configuration diagram representing a configuration example of a storage device according to an embodiment;

FIG. 2 shows an example of a connection configuration between a CPU and a memory as shown in FIG. 1;

FIG. 3 shows an example of setting the memory as shown in FIG. 1;

FIG. 4 shows an example of supplementary information about the memory setting as shown in FIG. 3;

FIG. 5 shows a configuration example of a hardware failure management table;

FIG. 6 shows a configuration example of a replacement target management table;

FIG. 7 shows a configuration of a cache directory;

FIG. 8 shows a configuration example of a reverse lookup table;

FIG. 9 shows an example of a free list;

FIG. 10 shows an example of a usage exclusion list;

FIG. 11 is a flowchart representing an example of an error processing procedure of the storage device according to the embodiment;

FIG. 12 is a flowchart representing an example of a specific segment exclusion processing procedure as shown in FIG. 11; and

FIG. 13 shows an example of a replacement recovery processing procedure.

DETAILED DESCRIPTION

Hereinbelow, preferred embodiments according to the present invention will be described in accordance with the accompanying drawings.

FIG. 1 is a system configuration diagram mainly representing a configuration example of a storage device 200 according to an embodiment. The storage device 200 is connected to a host computer (hereinafter abbreviated as a "host") 100 for data reading/writing operations with respect to the host 100. The storage device 200 formed of at least one unit of a computer implements an error processing method to be described later by running a control program 204A on the computer.

The storage device 200 includes multiple controllers, for example, controllers 200A and 200B. More than two controllers may be installed in the storage device 200. The controllers 200A and 200B have the respective functions duplexed, and are almost similarly configured. An explanation of the configuration of the controller 200B is, thus, omitted hereinafter except the one necessary to be described in terms of its relationship with the controller 200A.

The controller 200A includes a host I/F 201A, a CPU (Central Processing Unit) 202A, a memory 203A, a non-volatile memory 207A, a drive I/F 209A, and a drive unit 210A.

The host I/F **201A** is connected to the CPU **202A** via a signal line of PCI express standard, for example. As an interface with the host **100**, the host I/F **201A** connected to the host **100** via a fiber channel or Ethernet exchanges data with the host **100** using such protocol as iSCSI (Internet Small Computer System Interface). The host I/F **201A** receives a reading/writing request from the host **100**, interprets the content of the request, and delivers the request to the CPU **202A**.

In response to the request from the host **100** via the host I/F **201A**, the CPU **202A** controls data reading/writing operations with respect to a cache region **206A**, and the drive unit **210A** via the drive I/F **209A**. Specifically, the CPU **202A** performs data reading/writing operations with respect to the memory **203A** using Channels from “0” to “4”. The CPU **202A** controls data reading/writing operations by the control program **204A** between the drive units **210A** and **210B** via the drive I/F **209A** with reference to configuration information **208A** of the nonvolatile memory **207A** while accessing (a cache management region **205A**, and the cache region **206A** of) the memory **203A**.

The CPU **202A** includes a memory set unit (corresponding to a “memory set” in the drawing) **202A1**. The memory set unit **202A1** is a register for memory space management for the memory **203A**. The memory set unit **202A1** stores set information (refer to FIG. **3** and FIG. **4** to be described below) about memory setting to be described later. The set information about the memory set is stored in a nonvolatile manner in the nonvolatile memory **207A** as a part of the configuration information **208A**. The set information read from the nonvolatile memory **207A** is stored in the memory set unit **202A1**. The set information about the memory set indicates a range from the starting address to the ending address for defining each memory space in the memory **203A** with respect to the address space thereof (for example, each memory at Rank “0”, “1”, the cache region **206A**) from the side of the CPU **202A**.

The CPU **202A** is connected to a CPU **202B** of the other controller **200B** via a predetermined signal line. Occasionally, the CPU **202A** stores the request received from the host **100** via the host I/F **201A** in a cache region **206B** of the other controller **200B**. In this case, the CPU **202A** is allowed to order the CPU **202B** of the other controller **200B** to obtain and transfer the request and the like from the cache region **206B**, and to store the request and the like in the cache region **206A**.

The CPU **202A** constantly monitors occurrence of an error. Upon occurrence of an error, the CPU **202A** outputs an error number for identifying the error, a position where the error has occurred (the starting address and the ending address for defining the occurrence position in the memory **203A**), error information about occurrence of an uncorrectable error, and the number of occurrences of the uncorrectable error.

For example, the memory **203A** is configured at a dual Rank (Rank “0”, “1”) of DIMM (Dual Inline Memory Module) standard. The memory **203A** includes the control program **204A**, the cache management region **205A**, and the cache region **206A**. Under the control of the CPU **202A**, the control program **204A** controls data reading/writing operations with respect to the host **100** via the host I/F **201A**.

The cache region **206A** is a storage region to which multiple segments as an example of multiple management units are allocated to allow temporary storage of data in accordance with data reading/writing operations executed by the control program **204A** with respect to the host **100**. A segment number is given to each segment in the cache

region **206A**. The cache region **206A** allows storage of data in the segment as an allocation target under the control of the control program **204A**. Data are not stored in the specific segment excluded from the allocation target.

The control program **204A** as an example of a control unit controls data reading/writing operations under the control of the CPU **202A**. When an error occurs, it is determined whether or not the error occurrence position is in the cache region **206A**. If it is determined that the occurrence position is in the cache region **206A**, among the multiple segments, the specific segment having the occurrence position is excluded from the allocation target in the cache region **206A**. The segments remaining in the multiple segments are used for controlling the data reading/writing operations.

The control program **204A** determines whether or not the error is correctable. If it is determined that the error is not correctable, the specific segment is excluded from the allocation target.

The cache management region **205A** is a storage region for storing management information of the cache region **206A**. The management information includes each segment number of the multiple segments in the cache region **206A**, and includes the starting address to the ending address of each of those segments.

The cache management region **205A** manages a cache directory, a reverse lookup table, a free list, and a usage exclusion list, all of which are described later. The cache directory manages a correlation between multiple segments (storage unit) of the respective logical volumes (hereinafter simply abbreviated as a “volume”) for data storage, and the segment (management unit) in the cache region **206A**. A segment number is given to each segment of the respective volumes.

The reverse lookup table **205C** is a table for deriving the volume number of the volume for storing specific data, and the segment number in the volume, from the segment number of the segment in the cache region **206A** for storing the specific data.

The usage exclusion list manages identification information (segment number in the cache region) indicating the specific management unit excluded from the allocation target. The free list indicates the segment number of an unused segment in the cache region **206A**. Details of the cache directory, the reverse lookup table, the free list, and the usage exclusion list are described later.

The nonvolatile memory **207A** stores the configuration information **208A** in a nonvolatile manner. The configuration information **208A** includes the set information about setting of the memory, a hardware failure management table **211**, and a replacement target management table **212**, details of which are described later. The hardware failure management table **211** is a table for managing an error occurrence position in the cache region **206A**. The replacement target management table **212** is a table for managing the cache as a replacement target, which includes the cache region **206A** having the specific segment excluded from the allocation target. Details of the configuration information **208A** is described later.

The drive unit **210A** is constituted by at least a single drive unit, for example, an SSD (Solid State Drive), which enables nonvolatile data storage. In the embodiment, multiple drive units **210A** are used to constitute a RAID (Redundant Array of Independent Disks), for example. Alternatively, the drive unit **210A** may be formed as a magnetic disk device. In the embodiment, multiple drive units **210A** are used to constitute at least one volume. Each segment of the

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volume can store data as a target for reading/writing operations with respect to the host **100**.

The drive I/F **209A** is connected to the CPU **202A** via a signal line of PCI-Express (Peripheral Component Interconnect-Express) standard, for example. The drive I/F **209A** is an interface with the drive units **210A**, **210B**, and controls data reading/writing operations using such protocol as Serial Attached SCSI (Small Computer System Interface).

When the CPU **202A** receives a data writing request from the host **100** via the host I/F **201A**, the storage device **200** temporarily stores the data in the cache region **206A** as well as in the cache region **206B** of the other controller **200B** via the predetermined signal line, to manage the data by duplexing.

When the duplexing is completed, the CPU **202A** notifies the host **100** of completion of the writing operation to the cache region **206A** via the host I/F **201A**. The CPU **202A** stores the data temporarily stored in the cache region **206A** in the volume constituted by the drive units **210A**, **210B** as described later, and upon completion of the storage, further notifies the host **100** of completion of the writing operation to the volume via the host I/F **201A**.

When receiving a data reading request from the host **100** via the host I/F **201A**, the CPU **202A** reads the data from the volume, and temporarily stores the data in the cache region **206A**. The CPU **202A** supplies the data temporarily stored in the cache region **206A** to the host **100** via the host I/F **201A**.

FIG. **2** shows a configuration example of connection between the CPU **202A** and the memory **203A** as shown in FIG. **1**. As described above, the memory **203A** is of DIMM (Dual Inline Memory Module) standard with a dual Rank (Rank “0”, “1”) configuration. The CPU **202A** is connected to the memory **203A** at Rank **0** and Rank **1** through four channels so that data reading/writing operations are controlled with respect to the memory **203A** through the respective channels.

FIG. **3** shows an example of setting of the memory **203A** as shown in FIG. **1**. The set information related to setting of the memory **203A** partially constitutes the configuration information **208A**. A set value of item “Channel selection bit” is “6-7”. A set value of item “Rank selection bit” is “8”.

FIG. **4** shows an example of supplementary information about setting of the memory **203A** as shown in FIG. **3**. For example, a physical address is in a range from **0** to **40**. At the physical address “6-7”, data of “00” through the Channel “0”, “01” through the Channel “1”, “10” through the Channel “2”, and “11” through the Channel “3” are managed. The data “0” and “1” at the physical address “8” indicate the Rank “0” and “1”, respectively.

FIG. **5** shows a configuration example of the hardware failure management table **211**. The hardware failure management table **211** partially constitutes the configuration information **208A**. The hardware failure management table **211** manages an error number, an error occurrence address, information about whether an uncorrectable error occurs or not, and the number of occurrences of correctable errors, for each error which has occurred in the memory **203A**.

When an error occurs in the memory **203A**, the CPU **202A** registers the error number, the error occurrence address, the error information about occurrence of an uncorrectable error, and the number of occurrences of correctable errors in the hardware failure management table **211**.

FIG. **6** shows a configuration example of a replacement target management table **212**. As described above, the replacement target management table **212** partially constitutes the configuration information **208A**. The replacement

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target management table **212** represents a list of the replacement target caused by occurrence of error (hereinafter referred to as a “replacement target list”). The replacement target management table **212** indicates that the replacement target is of DIMM (memory **204A**) standard at Rank “1” through the Channel “0”, and a serial number is “0x12345678”. In the replacement target management table **212**, the replacement target is written by the CPU **202A** every time when the uncorrectable error occurs.

FIG. **7** shows a configuration of the cache directory **213**. The cache directory **213** is stored in the cache management region **205A**. The cache directory **213** manages a volume number for identifying each volume, a segment number in the volume, a segment number in the cache region **206A**, and an attribute. The cache directory **213** is a table for a reference to the attribute of the segment number in the cache region **206A** from the volume number, and the segment number in the volume. The cache directory **213** may be configured to manage the address corresponding to each of the segment numbers.

For example, the attribute includes “Clean” and “Dirty”. The attribute “Clean” represents that the data stored in the segment with the segment number in the volume with the volume number coincide with data stored in the segment with the segment number in the cache region **206A**. Meanwhile, the attribute “Dirty” represents that the data stored in the segment with the segment number in the volume with the volume number does not coincide with data stored in the segment with the segment number in the cache region **206A**.

The attribute “Clean” represents that the data stored in the segment with the segment number in the cache region **206A** coincide with data written into the drive unit **210A**. This means that no adverse influence is caused by loss of the data in the cache region **206A**. Meanwhile, the attribute “Dirty” represents that the data stored in the segment with the segment number in the cache region **206A** does not coincide with the data written into the drive unit **210A**. In this case, it is necessary to prevent loss of the data in the cache region **206A**. No data coincidence indicates that the data writing request from the host **100** has been received, but has not been reflected on the volume. However, the data have been duplexed between the controllers **200A** and **200B** like the case of the data coincidence. The attribute “-” represents that data are not stored in any of the segments in the cache region **206A**.

When the data are stored in the segment with the segment number “3” in the cache region **206A** corresponding to the segment number “0” in the volume with the volume number “0”, the attribute is “Clean”. This represents that the data stored in the segment with the segment number in the cache region **206A** coincide with the data written into the drive unit **210A**. Referring to FIG. **7**, all entered data are not always text data, but may have the tree or hash structure.

FIG. **8** shows a configuration example of the reverse lookup table **205C**. The reverse lookup table **205C** is stored in the cache management region **205A**. As described above, the reverse lookup table **205C** is a table for deriving the volume number of the volume for storing the data, and the segment number in the volume from the segment number of the segment in the cache region **206A** for storing the data. That is, making reference to the reverse lookup table **205C** allows reverse lookup of the cache directory **213**.

The reverse lookup table **205C** manages the segment number in the cache region **206A**, the volume number, and the segment number in the volume. Explanations of the segment number in the cache region **206A**, the volume

number, and the segment number in the volume are omitted because of similarity to those described referring to FIG. 8.

FIG. 9 shows an example of the free list 205D. The free list 205D is stored in the cache management region 205A, and updated by the control program 204A under the control of the CPU 202A. As described above, the free list 205D indicates the segment number of an unused segment in the cache region 206A. The example in the drawing represents that the segments with numbers "0" and "5" are not in use.

FIG. 10 shows an example of the usage exclusion list 205E. The usage exclusion list 205E is stored in the cache management region 205A, and updated by the control program 204A under the control of the CPU 202A. The example in the drawing represents that the segment with the segment number "2" in the cache region 206A is a usage exclusion target.

The storage device 200 according to the embodiment is configured as above. An example of an error processing method to be implemented in the storage device 200 is described. The error processing method for a storage device 200 having multiple controllers 200A, 200B provided with CPUs 202A, 202B for controlling data reading and writing operations with respect to at least one host computer 100 includes: a management unit allocation step for causing the CPU 202A, 202B to allocate multiple segments capable of temporarily storing the data to each of the cache regions 206A, 206B of the controllers 200A, 200B in accordance with the data reading and writing operations under the control of the control program 204A; a determination step for causing the CPU 202A to determine, when an error occurs, whether or not an occurrence position of the error is in the cache region 206A under the control of the control program 204A; and a control step for causing the CPU 202A, when it is determined that the occurrence position is in the cache region 206A, to exclude a specific segment including the occurrence position among the multiple segments from the allocation target in the cache region 206A, and to control the data reading and writing operations using segments remaining in the multiple segments.

FIG. 11 is a flowchart representing an example of an error processing procedure in the storage device 200 according to the embodiment. The control program 204A is caused to execute the error processing under the control of the CPU 202A. The following description, however, is simplified on the assumption that the control program 204A executes the error processing. The controller 200A is mainly described as follows except that the controller 200B involves the specific processing.

The free list 205D manages the segment number of the segment in which data are not stored in the cache management region 205A. The control program 204A uses the respective segments in the cache management region 205A with reference to the free list 205D.

The storage device 200 causes the control program 204A of the controller 200A to constantly detect an error. When an error occurs, it is determined whether or not the error has occurred in the cache region 206A. If the error has occurred in the cache region 206A, the program obtains an error No., information about whether or not the error which has occurred is correctable, and an error occurrence address corresponding to a position at which the error has occurred.

In step S10, when an error occurs, the control program 204A obtains an error occurrence address from the hardware failure management table 211. In step S20, the control program 204A determines whether or not the error is correctable. If it is determined that the error is not correctable,

the control program 204A executes step S30. If it is determined that the error is correctable, step S50 is executed as described later.

In step S30, the control program 204A determines whether or not continuous usage is possible. Specifically, if the error is correctable, and the number of error occurrences in the segment with the address which includes the occurrence address is equal to or smaller than a threshold value, the control program 204A determines that the continuous usage is possible.

If it is determined that the continuous usage is possible, the control program 204A executes step S40. If it is determined that the continuous usage is not possible, step S50 is executed as described later.

In step S40, the control program 204A records the error number indicating occurrence of the error in the hardware failure management table 211. The error processing is then finished.

Meanwhile, in step S50, the control program 204A determines whether or not the error occurrence position is in the cache region 206A. If it is determined that the error occurrence position is not in the cache region 206A, the control program 204A executes a controller block processing. In the controller block processing, the control program 204A blocks the controller (the controller 200A in this example) having the error occurred (step S60).

Meanwhile, if it is determined that the error occurrence position is in the cache region 206A, the control program 204A determines that there is a locality in the error occurrence position (step S70). Specifically, in the case where errors have occurred at multiple positions in the cache region 206A, the control program 204A determines whether or not the error occurrence positions are concentrated in the specific area of the cache region 206A.

If it is determined that there is no locality, the control program 204A executes the controller block processing. If it is determined that locality exists, the segment number in the cache region 206A is identified with reference to the usage exclusion list 205E (step S80).

The control program 204A executes a segment exclusion processing (step S90). Specifically, the control program 204A adds the segment number of the segment to the usage exclusion list 205E (refer to FIG. 10) as a part of the configuration information 208A. When the segment number of the segment is added to the usage exclusion list 205E, the control program 204A no longer uses the segment with the segment number. Details of the segment exclusion processing is described later.

The control program 204A executes an error occurrence memory identification processing (step S100) to identify the memory having the error. Specifically, the control program 204A constantly obtains the error information. Based on the error information, the memory having the error (the memory 203A in this case) is identified.

The control program 204A registers the memory 203A in the replacement target management table 212 (refer to FIG. 6) (step S110).

FIG. 12 is a flowchart representing a specific example of a segment exclusion processing procedure as shown in FIG. 11. In step S21, the control program 204A determines whether or not the segment is in use.

Specifically, the control program 204A determines whether or not the segment is in use with reference to the free list 205D. When the segment is not in use, the control program 204A executes step S27.

When it is determined that the segment is in use, the control program 204A identifies the volume number of the

volume using the cache region **206A** having the error occurrence position, and the segment number in the volume with reference to the reverse lookup table **205C** (step **S22**).

The control program **204A** confirms the attribute corresponding to the volume number with reference to the cache directory **213** (step **S23**). If the attribute is "Clean", the control program **204A** executes step **S25** to be described later. If the attribute is "Dirty", step **S24** is executed.

In step **S24**, the control program **204A** sends a request to the other controller **200B** to reflect data of the segment in the volume. Then in step **S25**, the control program **204A** changes the entry into the cache directory **213** of the own controller **200A** to "-" (without caching).

In step **S26**, the control program **204A** deletes a reference destination of the reverse lookup table **205C**. The control program **204A** further adds the segment number in the cache region **206A** to the usage exclusion list **205E**.

FIG. **13** shows an example of a replacement recovery processing procedure. In step **S31**, the control program **204A** refers to a serial number of the memory **203A** registered as a replacement target list of the replacement target management table **212**.

The control program **204A** determines with respect to coincidence with the serial number of the replacement target management table **212**. If non-coincidence is determined (for example, the memory **203A** has been replaced), the control program **204A** deletes an address group corresponding to the replaced memory **203A** from the error occurrence address list (step **S33**).

If coincidence is determined (for example, the memory **203A** has not been replaced), the control program **204A** executes step **S34**. In step **S34**, the control program **204A** extracts an address group corresponding to the memory **203A** from the error occurrence address list.

In step **S35**, the control program **204A** calculates the segment number in the cache region **206A**, which corresponds to each address. In step **S36**, the control program **204A** adds the calculated segment number to the usage exclusion list **205E**.

The storage device **200** according to the embodiment includes multiple controllers **200A**, **200B** for controlling data reading and writing operations with respect to at least one host computer. The controllers **200A**, **200B** include cache regions **206A**, **206B** to which multiple segments (management units) capable of temporarily storing the data are allocated in accordance with the data reading and writing operations, and a control unit (CPU **202A**, control program **204A**) for controlling the data reading and writing operations. When an error occurs, the control unit determines whether or not an occurrence position of the error is in the cache region **206A**. When it is determined that the occurrence position of the error is in the cache region **206A**, the control unit excludes, among the multiple segments, a specific segment that contains the occurrence position of the error from an allocation target in the cache region **206A** to control the data reading and writing operations using segments remaining in the multiple segments.

Even if an error occurs in the cache region **206A**, the configuration ensures to avoid blocking of the controller having the error as much as possible. Even if an error occurs in the cache region **206A**, the controller having the error can be continuously operated. This allows the storage device **200** to continue data reading/writing operations by the controllers **200A**, **200B** to the utmost. In the case where the other controller has an error, and is blocked, the frequency of system going down as the whole storage device **200** can be reduced.

In the embodiment, the control program **204A** determines whether or not the error which has occurred in the cache region **206A** is correctable under the control of the CPU **202A**. If the error is not correctable, the specific segment is excluded from the allocation target. This ensures to prevent blocking of the controller including the cache region **206A** having the correctable error, and to attain stable operation of the storage device **200** as a whole.

In this embodiment, each of the controllers **200A**, **200B** includes the usage exclusion list **205E** for managing the segment number indicating the specific segment excluded from the allocation target. This ensures to control the control program **204A** not to use the segment with the segment number managed by the usage exclusion list **205E** in the cache region **206A** by mistake.

In this embodiment, each of the controllers **200A**, **200B** includes the cache directory **213** for managing a correlation between the multiple segments in the volume for storing data and the multiple segments in the cache region **206A** for storing the data. This allows the control program **204A** to easily grasp the correlation between the segment in the volume having the data stored, and the segment in the cache region **206A** having the data stored.

In this embodiment, each of the controllers **200A**, **200B** includes the hardware failure management table **211** for managing the error occurrence position in the cache region **206A**. This allows easy recognition of the hardware to be replaced only by referring to the hardware failure management table **211**.

In this embodiment, each of the controllers **200A**, **200B** includes the replacement target management table **212** for managing the cache as the replacement target, which includes the cache region **206A** having the specific segment excluded from the allocation target. This allows easy recognition of the cache to be replaced only by referring to the hardware failure management table **211**.

The present invention is not limited to the embodiment as described above, and includes various modifications and equivalent configurations within the scope of the appended claims. For example, the above-described embodiments have been described in detail in order to facilitate the understanding of the present invention, and the present invention is not necessarily limited to those including all of the described configurations. The respective elements described in parallel in the embodiment may be interpreted that at least one of the elements is connected to other elements in series.

The present invention is applicable to the storage device which relates to the technology for avoiding blocking of the controller as a whole despite the error which occurs in the cache region of the storage device.

What is claimed is:

1. A storage device comprising a plurality of controllers for controlling data reading and writing operations with respect to at least one host computer, wherein

each of the controllers includes:

a cache region to which a plurality of management units capable of temporarily storing the data are allocated in accordance with the data reading and writing operations; and

a control unit for controlling the data reading and writing operations, and when an error occurs, the control unit determines whether or not an occurrence position of the error is in the cache region, and when it is determined that the occurrence position is in the cache region, the control unit excludes, among the plural management units, a specific management unit that contains the

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occurrence position from an allocation target in the cache region, to control the data reading and writing operations using remaining management units in the plural management units.

2. The storage device according to claim 1, wherein the control unit determines whether or not the error is correctable, and, when the error is not correctable, the control unit excludes the specific management unit from the allocation target.

3. The storage device according to claim 2, wherein each of the controllers includes a usage exclusion list for managing identification information indicating the specific management unit excluded from the allocation target.

4. The storage device according to claim 1, wherein each of the controllers includes a cache directory for managing a correlation between plural storage units in a volume which stores the data, and the plural management units in the cache region which stores the data.

5. The storage device according to claim 1, wherein each of the controllers includes a hardware failure management table for managing the occurrence position in the cache region.

6. The storage device according to claim 1, wherein each of the controllers includes a replacement target management table for managing a cache as a replacement target, that includes the cache region having the specific management unit excluded from the allocation target.

7. An error processing method for a storage device having a plurality of controllers each provided with a control unit for controlling data reading and writing operations with respect to at least one host computer, the error processing method comprising:

a management unit allocation step for causing the control unit to allocate a plurality of management units capable of temporarily storing the data to each cache region of the controllers in accordance with the data reading and writing operations;

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a determination step for causing the control unit to determine, when an error occurs, whether or not an occurrence position of the error is in the cache region; and

a control step for causing the control unit, when it is determined that the occurrence position is in the cache region, to exclude a specific management unit including the occurrence position among the management units from the allocation target in the cache region, and to control the data reading and writing operations using remaining management units in the plural management units.

8. The error processing method according to claim 7, wherein in the management unit allocation step, the control unit determines whether or not the error is correctable, and, when the error is not correctable, excludes the specific management unit from the allocation target.

9. The error processing method according to claim 8, wherein

each of the controllers includes a usage exclusion list for managing identification information indicating the specific management unit excluded from the allocation target, and

in the control step, the control unit controls the data reading and writing operations so as not to use the specific management unit corresponding to identification information managed in the usage exclusion list.

10. The error processing method according to claim 7, wherein

each of the controllers includes a cache directory for managing a correlation between a plurality of storage units in a volume which stores the data, and the plural management units in the cache region which stores the data, and

in the control step, the control unit identifies a predetermined storage unit in the volume corresponding to a predetermined management unit in the cache region with reference to the cache directory.

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